

A Additional Qualitative Samples

Numerically, previous findings [44] have shown it is challenging to see the difference due to the nature of automated text scoring criteria. Thus, we include sample image-caption pairs showing the qualitative improvement in image caption as ViZer-trained VLMs display a more accurate description of an image with minimal garbage. We include SmolVLM-Base [29] in Figure 5 and Figure 6, and Qwen2-VL [39] results in Figure 7 and Figure 8.

A.1 Vision Feature Space Visualizations

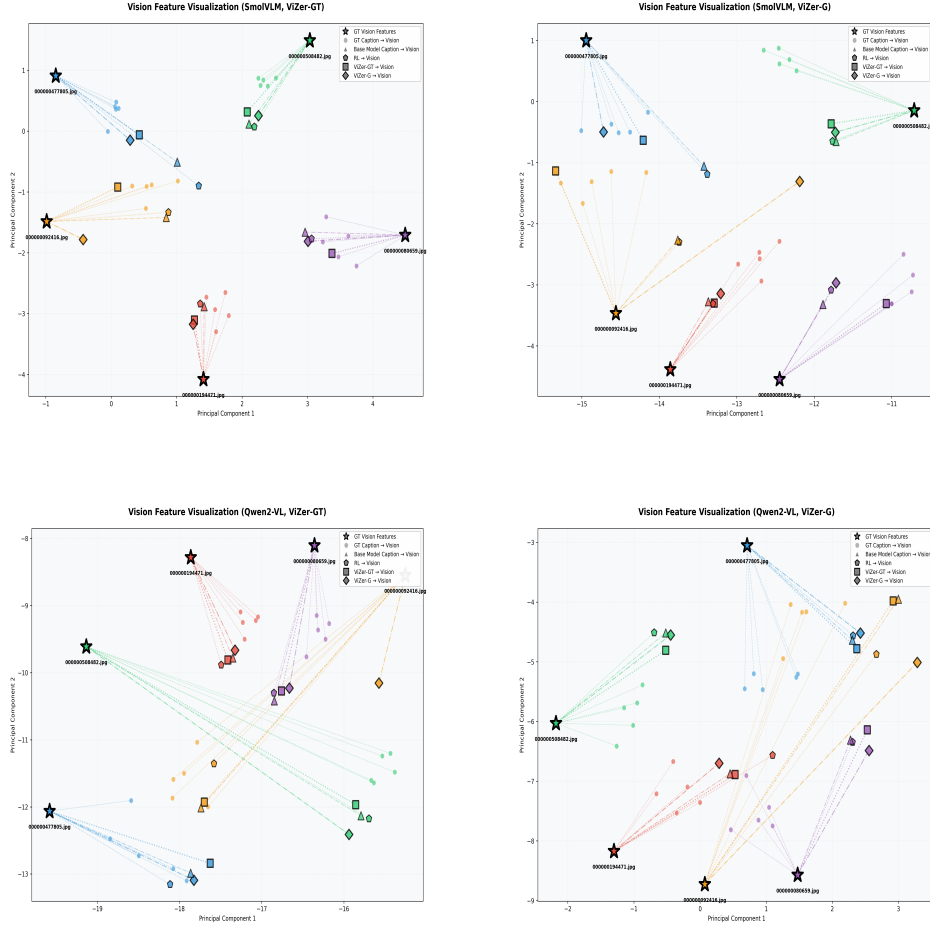


Figure 4: Visual Feature Comparisons on post-training generated image captions for SmolVLM-Base [29] (top) and Qwen2-VL [39] (bottom) with ViZer_{GT} (left) and ViZer_G (right). 2-dimensional PCA is applied to text and vision features for ease of comparison. Points closer to stars represent similar latents.

To better understand the impact of ViZer on vision-language models (VLMs), we visualize the 2-dimensional PCA projections of visual features for a sample image, its ground-truth captions, the original generated captions, and the post-ViZer generated captions in Figure 4. The visualization demonstrates that ViZer generally brings the generated caption embeddings closer to the corresponding visual features, indicating improved semantic alignment between modalities. However, the features are not perfectly overlapping, which reflects a key property of our approach: while ViZer enhances grounding, it avoids forcing captions to perfectly match visual embeddings. This balance is important because no textual description can fully capture all aspects of an image, and overly aggressive alignment may lead to unnecessarily verbose or unnatural captions.

B Evaluation Metrics

B.1 BLEU

The BLEU [31] (Bilingual Evaluation Understudy) score measures the similarity between a generated sentence $\hat{y}$ and one or more reference sentences y . It is based on modified n -gram precision, combined using a weighted geometric mean and adjusted by a brevity penalty (BP) to avoid favoring shorter outputs. Formally:

$$\text{BLEU} = \text{BP} \cdot \exp \left(\sum_{n=1}^N w_n \ln p_n \right), \quad \text{BP} = \exp \left(-\max \left(0, \frac{r}{c} - 1 \right) \right),$$

$$p_n = \frac{\sum_{s \in G_n(\hat{y})} \min(C(s, \hat{y}), \max_{y \in \{y^{(i)}\}} C(s, y))}{\sum_{s \in G_n(\hat{y})} C(s, \hat{y})},$$

where $G_n(\cdot)$ is the set of n -grams, $C(s, \cdot)$ is the count of s , c is the candidate length, r is the effective reference length, and w_n are typically uniform weights over $n = 1, \dots, 4$.

B.2 ROUGE

ROUGE [24] (Recall-Oriented Understudy for Gisting Evaluation) measures the overlap of n -grams between a candidate and reference sentence. For ROUGE-N, recall, precision, and F1 are defined as:

$$\text{F1} = \frac{2 \cdot \text{P} \cdot \text{R}}{\text{P} + \text{R}}, \quad \text{R} = \frac{|\text{overlapping } n\text{-grams}|}{\text{total } n\text{-grams in reference}}, \quad \text{P} = \frac{|\text{overlapping } n\text{-grams}|}{\text{total } n\text{-grams in candidate}}.$$

Variants like ROUGE-L rely on the longest common subsequence (LCS).

B.3 CIDEr

The CIDEr [38] (Consensus-Based Image Description Evaluation) metric evaluates the similarity of a generated caption to a set of human references by leveraging TF-IDF weighted n -gram statistics. Let g_n and r_n be the TF-IDF vectors for the candidate and reference captions, respectively. The score is computed as:

$$\text{CIDEr} = \frac{1}{N} \sum_{n=1}^N \frac{g_n \cdot r_n}{\|g_n\| \cdot \|r_n\|},$$

where N denotes the number of n -gram orders considered (commonly $N = 4$). CIDEr rewards captions that align closely with human consensus while emphasizing informative, distinctive n -grams.

B.4 BERTScore

BERTScore [47] evaluates the semantic similarity between candidate and reference captions by comparing contextual embeddings from pretrained language models such as BERT. Given embeddings $\mathbf{e}_c$ for candidate tokens and $\mathbf{e}_r$ for reference tokens, the precision, recall, and F1 are computed as:

$$F_1 = \frac{2 \cdot P \cdot R}{P + R}, \quad P = \frac{1}{|C|} \sum_{c \in C} \max_{r \in R} \cos(\mathbf{e}_c, \mathbf{e}_r), \quad R = \frac{1}{|R|} \sum_{r \in R} \max_{c \in C} \cos(\mathbf{e}_r, \mathbf{e}_c),$$

where $\cos(\cdot, \cdot)$ measures cosine similarity between token embeddings. Unlike BLEU and ROUGE, BERTScore captures semantic equivalence beyond exact n -gram overlaps and correlates better with human judgment.

B.5 CLIPScore

CLIPScore [13] leverages pretrained CLIP (Contrastive Language–Image Pretraining) models to directly measure the alignment between a generated caption $\hat{y}$ and the corresponding image I . Unlike text-only metrics such as BLEU or ROUGE, CLIPScore evaluates captions in a multimodal embedding space, capturing whether the caption is semantically consistent with the visual content.

Formally, let $\mathbf{e}_I = f_{\text{img}}(I)$ and $\mathbf{e}_{\hat{y}} = f_{\text{text}}(\hat{y})$ be the normalized embeddings of the image and candidate caption from CLIP’s image and text encoders. The score is computed as the cosine similarity:

$$\text{CLIPScore}(I, \hat{y}) = \cos(\mathbf{e}_I, \mathbf{e}_{\hat{y}}) = \frac{\mathbf{e}_I \cdot \mathbf{e}_{\hat{y}}}{\|\mathbf{e}_I\| \|\mathbf{e}_{\hat{y}}\|}.$$

Variants include length-penalized versions that discourage overly short captions and normalized versions that map scores into $[0, 1]$. CLIPScore has been shown to better correlate with human judgments of caption quality compared to traditional n -gram-based metrics, since it evaluates semantic faithfulness to the image rather than just textual overlap with references.

C Limitation

Currently, ViZer is limited to image captioning tasks, as extending the visual feature alignment to visual question answering (VQA) remains a challenging endeavor. In VQA, models often focus on localized regions or specific objects rather than the entire visual scene, making direct alignment between answers and global visual features a non-trivial task. In future work, we aim to design a scheme that enables ViZer to integrate VQA answer representations with corresponding visual semantics effectively.

Another limitation lies in the lack of suitable automated evaluation metrics for self-supervised VQA settings. Existing metrics either depend heavily on human-annotated references or rely on large language models, which may introduce additional biases. We plan to develop a reference-free, non-LLM-dependent scoring mechanism to more accurately evaluate alignment quality and enhance the reliability of performance assessment.

Finally, we envision integrating ViZer into a continuous learning loop with the underlying VLM. This would allow ViZer to evolve from an alignment module into a powerful pre-training strategy, enabling models to improve their multimodal grounding over time without requiring extensive manual supervision.



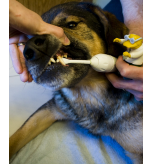
GT: A woman in a bikini riding a wave on a surfboard.
Base: <PERSON> in 2008
RL: <PERSON> in 2008
ViZer_{GT}: Woman surfing in the ocean.
ViZer_G: Surfer in a bikini riding a wave



GT: A cat playing with it's reflection in a mirror.
Base: My cat is so cute.
RL: My cat has a mirror fetish.
ViZer_{GT}: A cat looking at herself in a mirror.
ViZer_G: A cat's paw in a mirror



GT: a close up of a person wearing a suit and tie
Base: Me in 2010
RL: A man in a suit and hat on a train.
ViZer_{GT}: A man in a suit and hat on a train
ViZer_G: A man in a suit and hat on a train.



GT: A dog having his teeth brushed with a novelty electric toothbrush.
Base: A dog getting a dental cleaning.
RL: A dog getting a dental cleaning.
ViZer_{GT}: A dog getting his teeth brushed by his owner.
ViZer_G: A dog getting his teeth brushed with an electric toothbrush



GT: A man holding a stick standing next to a green hillside.
Base: <PERSON> at the start of the trail
RL: <PERSON> and <PERSON> on the trail
ViZer_{GT}: A man with a backpack and trekking poles on the trail
ViZer_G: A man crosses a stream on the way to the temple.



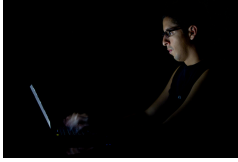
GT: A toddler in the bathroom brushing his teeth
Base: In this image we can see a child brushing his teeth.
RL: In this image we can see a child brushing his teeth with a toothbrush.
ViZer_{GT}: little boy brushing his teeth in the bathroom
ViZer_G: a little boy brushing his teeth in the bathroom



GT: A pile of apples sitting next to a pile of green apples.
Base: In this image I can see many fruits on the table.
RL: In this image I can see many fruits on the table.
ViZer_{GT}: In this image we can see a variety of fruits at a fruit stand
ViZer_G: In this image we can see a pile of fruits.



GT: A view of a TV and a statue laying on the ground next to a window.
Base: The living room in 2009
RL: The living room in 2009
ViZer_{GT}: A mannequin in front of a television set.
ViZer_G: A mannequin on a cart in the living room.



GT: A man with glasses sitting in front of a laptop computer.
Base: ITAP of a guy working in the dark
RL: ITAP of a guy working on his laptop in the dark
ViZer_{GT}: A man using a laptop in the dark.
ViZer_G: A man using a laptop computer in the dark



GT: The open doors looking into a subway from the station.
Base: Interior of a TGV
RL: Interior of a TGV
ViZer_{GT}: Interior of a train
ViZer_G: Interior of a train at the station



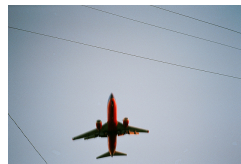
GT: a man is swinging a baseball bat at a game
Base: Ryan Zimmerman of the Washington Nationals
RL: Ryan Zimmerman of the Washington Nationals in 2007
ViZer_{GT}: Washington Nationals player batting during a game
ViZer_G: Washington Nationals' pitcher Stephen Strasburg (right) throws a pitch during the first inning of a game against the Washington Nationals at Nationals Park on April 19 2



GT: There is a dog holding a Frisbee in its mouth.
Base: My doggo loves the beach!
RL: My doggo loves the beach!
ViZer_{GT}: My dog playing fetch on the beach.
ViZer_G: A pupper on the beach.



GT: A bathroom with a white toilet and a sink next to a shower.
Base: Bathroom with orange shower curtain.
RL: Bathroom with orange shower curtain.
ViZer_{GT}: A bathroom with a shower and sink.
ViZer_G: Bathroom with shower curtain and sink.



GT: A colorful plane flying over head and telephone wires
Base: ITAP of an airplane
RL: ITAP of an airplane
ViZer_{GT}: ITAP of an airplane flying in the sky
ViZer_G: ITAP of an airplane flying over power lines

Figure 5: Qualitative captioning comparison with SmolVLM-Base [29] for sample images from OpenImage [17] dataset. We compare ground-truth, baseline, reinforcement-learning, ViZer_{GT}, and ViZer_G generated captions. Results continue on Figure 6

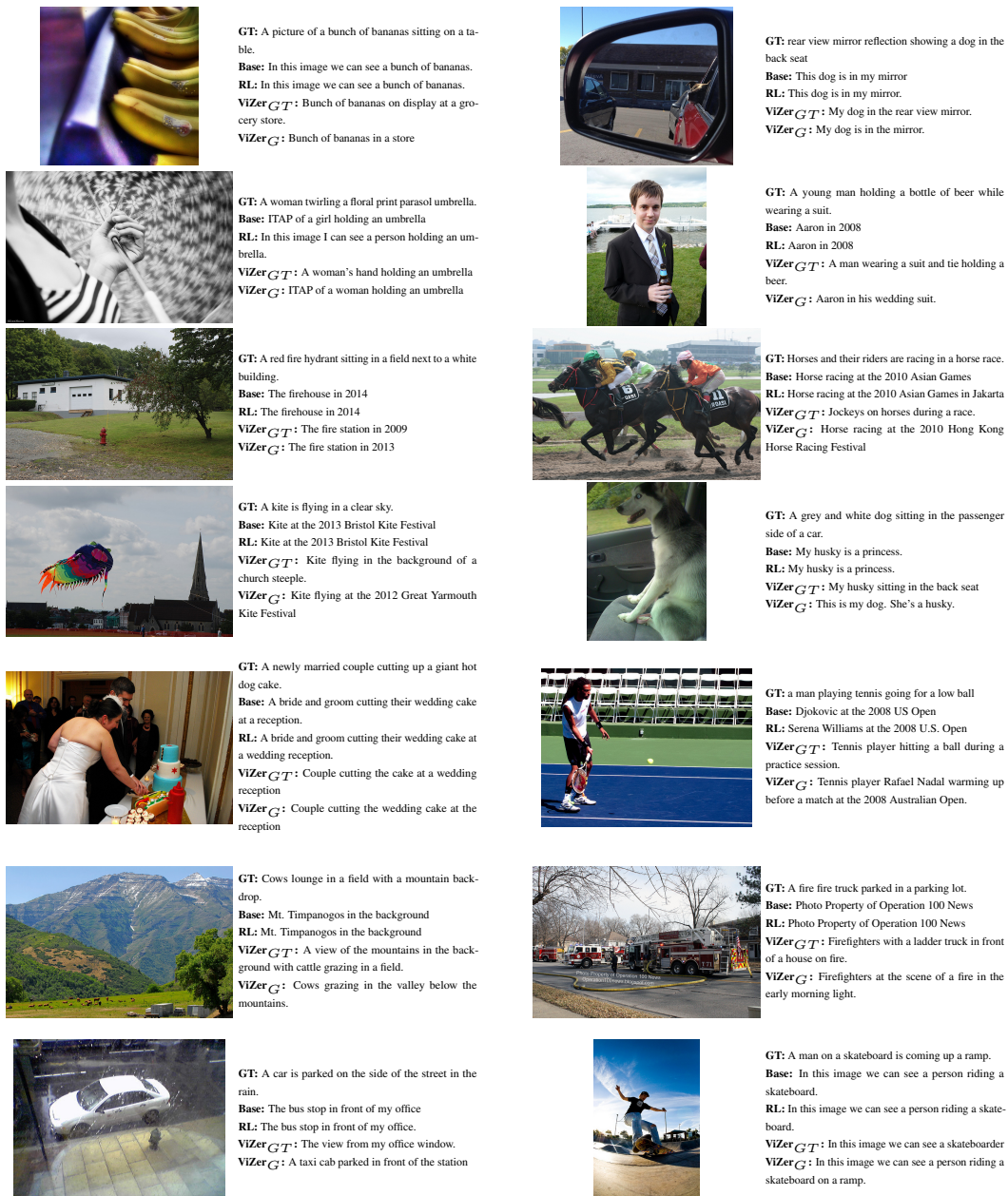


Figure 6: Continuation of Figure 5. Generated caption samples for the OpenImage dataset.

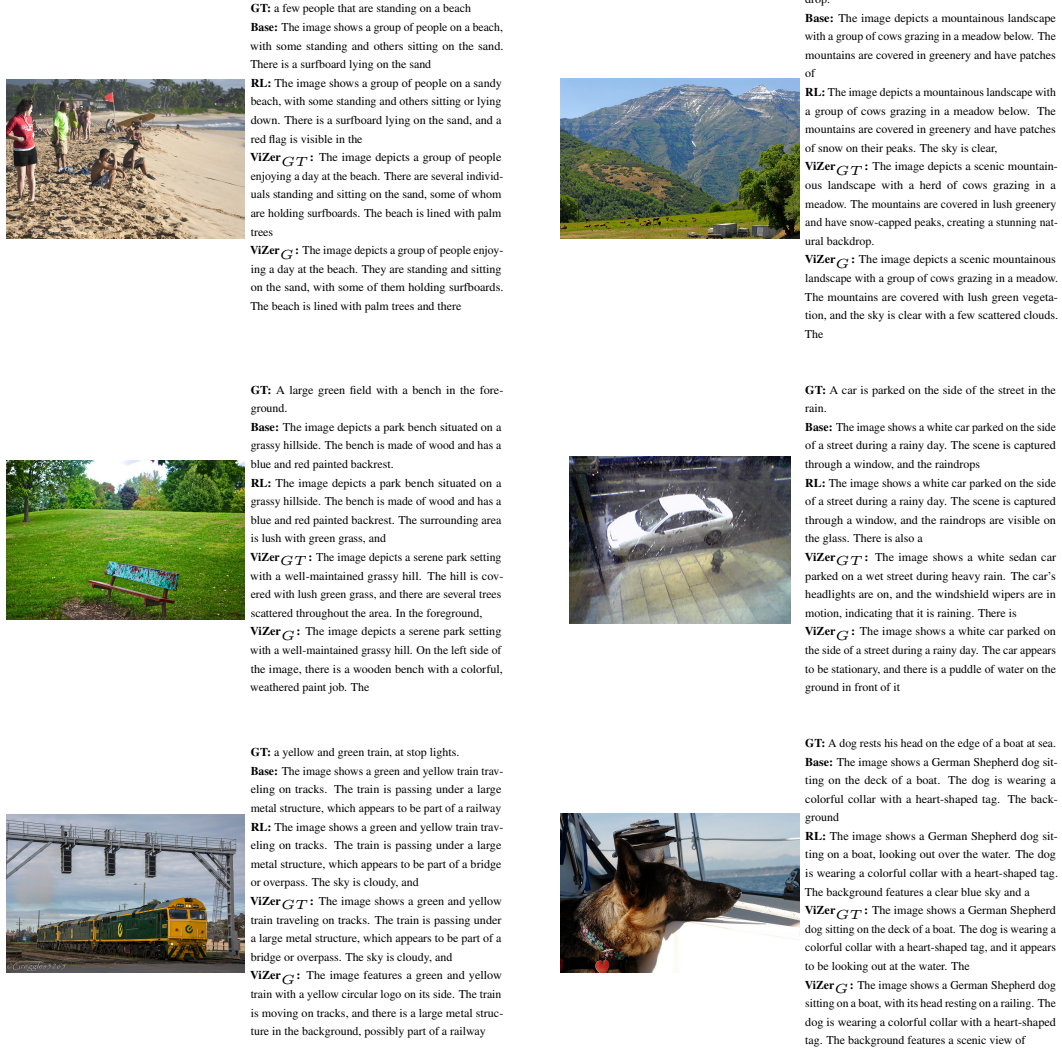


Figure 7: Qualitative captioning comparison with Qwen2-VL [39] for sample images from OpenImage [17] dataset. We compare ground-truth, baseline, reinforcement-learning, ViZer_{GT}, and ViZer_G generated captions. Results continued in Figure 8

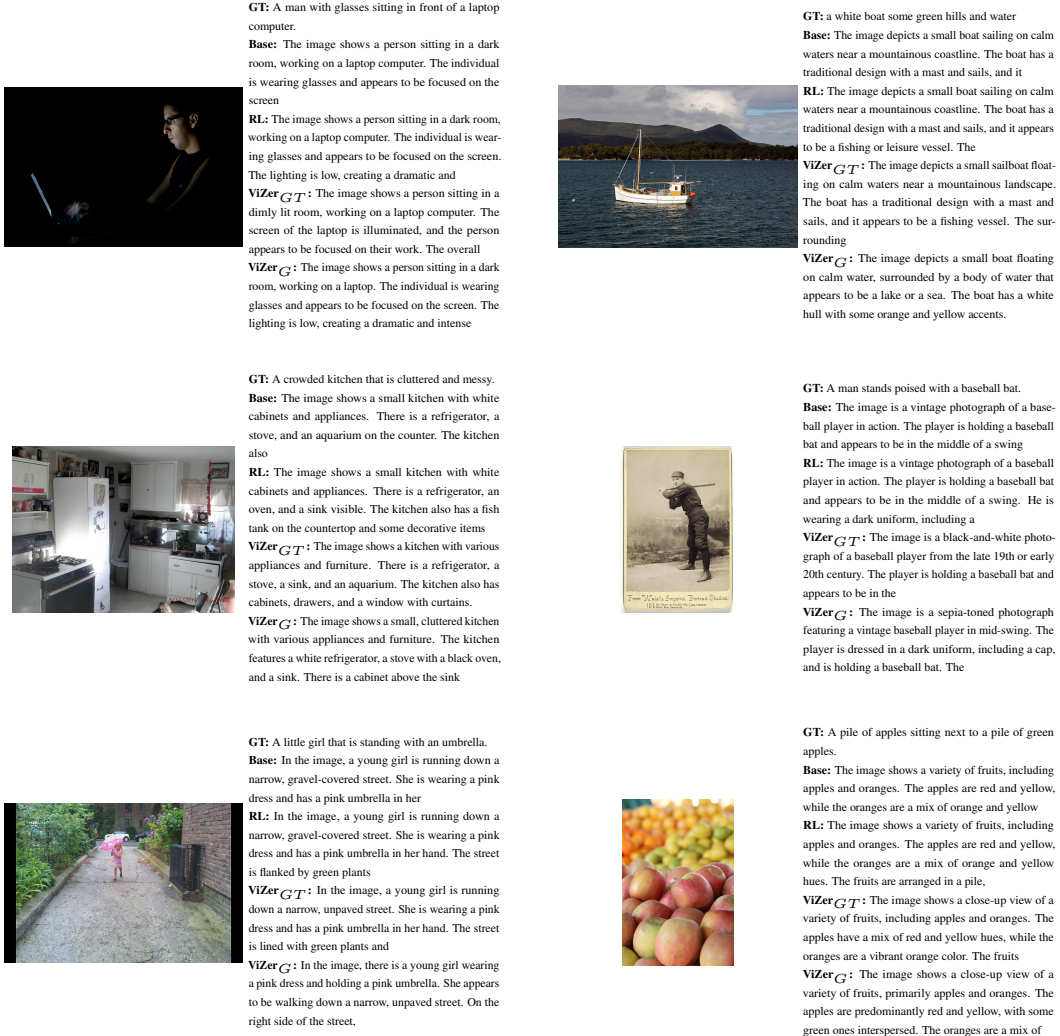


Figure 8: Continuation of Figure 7. Generated caption samples for the OpenImage dataset.